

Advanced Digital Design 130323

VHDL Synthesis – Simple Counter Synthesis (Assignment 5)

Date: Tuesday, November 26, 2024

Submission:	By not more than 2 people together
Deadline: Preparation work	Tuesday, December 3, 2024 (17:15) – via moodle
Deadline: Final report	Tuesday, December 17, 2024, (17:15) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

What is this lab about?

- The purpose of this exercise is to walk you through Precision VHDL synthesizer and be familiar with the tool.

What is this lab NOT about?

There are many things that this lab is not... in particular...

- The lab is not yet intended to thoroughly exercise you with the VHDL language.
- The lab is not yet intended to exercise you with Xilinx FPGA technology.

A. Preparation work

Note: Preparation work has an earlier deadline and is considered as an independent separate assignment.

This assignment is based on a VHDL file (`cntr.vhd`) given for this assignment's directory. Look into this file and answer the following questions. Each answer must include the portion of the code that supports the answer.

1. What is the total number of inputs and outputs that this design has?
2. Which constructs in the counter's VHDL code indicate that the design has Flip-Flops?
3. What is the total number of Flip-Flops that this design has?
4. Is the TC generated directly by a Flip-Flop or does it require combinational logic?

And...

5. Copy `cntr.vhd` to `cntr64.vhd` and change the design in the new file in such that it will be a **64-bit** counter. You will use the new file **only for step F** of this assignment (you must submit `cntr64.vhd` as part of your preparation work).

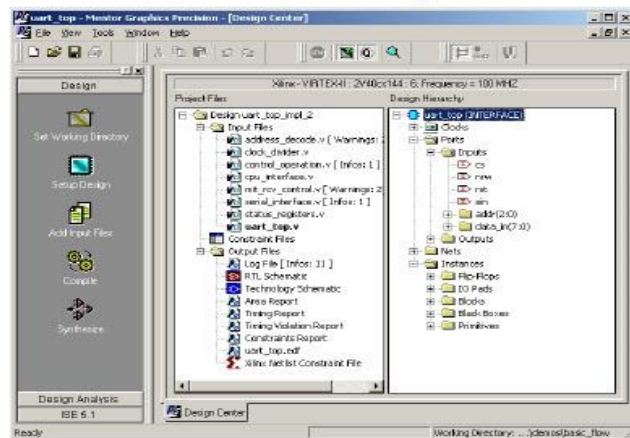
B. Running Precision

Note: The exercise directory includes a PDF file with a step-by-step tutorial explaining how to run Precision. Follow the step-by-step instructions given in that file.

Yet, be aware of the differences between that file and specific technology details given below.

In short, Precision's operation via the **GUI** (Graphical User Interface) is directed by a set of icons located in the “**design bar**” at the left of the screen. The designer is requested to:

The Precision Design Center



- Set a project (Create a new project or open an existing one).
- Setup Design (Define target technology and frequency)
- Add input files
- Compile
- Synthesize

Precision's **Design Center** provides an easy way to look into the project's **input files** (VHDL sources) and **output files** (results and reports).

Precision's **transcript** reflects each GUI meaningful operation in a Precision **TCL** (Tool Command Language) command and show the response of Precision to that command. (The complete set of commands and responses is called the “**log file**”, it is one of the synthesis important files)

- More related help about precision can be found in the “on line help” of the tool.

C. More Details About the Target Technology (Xilinx Artix-7 FPGA)

- The target technology selected for our design is Xilinx Artix-7. Technical details about this technology can be found in Xilinx' datasheet. Given along with the exercise
- Further details about Xilinx technology can be found in <http://www.xilinx.com>
- In this lab, all you need to be aware are the family table and the package table given in the datasheet

D. Files and Directories

- Open a new directory in which you have read and write permissions. Copy `cntr.vhd` file into that new directory.

E. Constraints and Target technology

Note: See the step by step tutorial for further advice – **be aware of specific changes required for this exercise**

- Target to the following Xilinx device:
 - Family: Artix-7
 - Device: 15T
 - Package: CPG236
 - Speed: -3
- Clock frequency: 100 MHz
- Make sure that you provide 0 ns input and output delays.

F. 64-bit Counter

- Open a new directory and repeat the synthesis flow in for the 64-bit counter (that was prepared in A.5).

G. What Must be Included in Your Report:

Note: When including reports and logs from the synthesis tool in a Microsoft Word document, you need to select Courier New fonts for that report/log section. Otherwise, the report/log is not convenient for reading

(Please select a reasonable font size for that section, we are not indicating what is reasonable, you should make your own choice).

For each of the counters (4-bit and 64-bit), the following items shall be included in your report:

- Synthesis log file¹
- Timing report
- Area report
- RTL view drawing (single page – i.e., don't use the multipage drawing option)
- Technology view drawing, single page² with LUTs presented as logic³.

¹This is a file that contains everything that you have done with the synthesis tool and how the tool have responded to – look for this file or copy and paste the transcript

²In the Technology Schematic, place cursor on the drawing, right click with mouse and unselect the “Multipage drawing”

³In the Technology Schematic, place cursor on the drawing, right click with the mouse and select “Show LUT information”